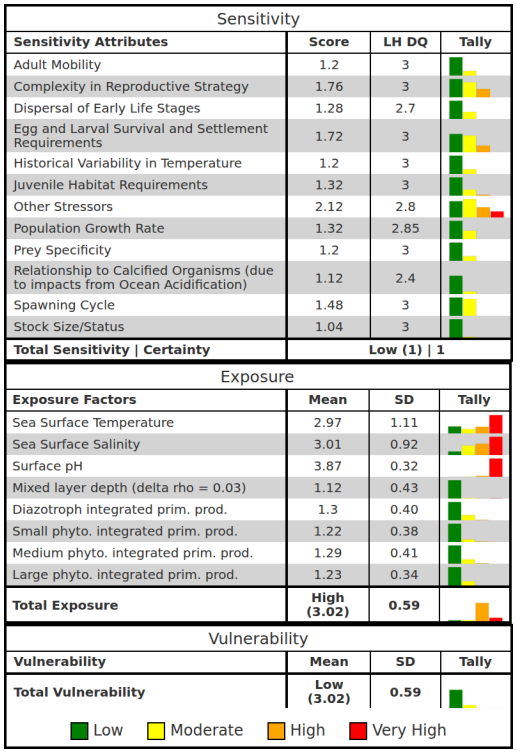
Butterfish (*Peprilus triacanthus*)

Summary Scores

- **Sensitivity:** Low
- **Exposure:** High
- **Vulnerability:** Low



Sensitivity

Sensitivity Attributes	Score	LH DQ
Adult Mobility	1.2	3
Complexity in Reproductive Strategy	1.76	3
Dispersal of Early Life Stages	1.28	2.7
Egg and Larval Survival and Settlement Requirements	1.72	3
Historical Variability in Temperature	1.2	3
Juvenile Habitat Requirements	1.32	3
Other Stressors	2.12	2.8
Population Growth Rate	1.32	2.85
Prey Specificity	1.2	3
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.12	2.4
Spawning Cycle	1.48	3
Stock Size/Status	1.04	3
Total Sensitivity | Certainty	**Low (1) | 1**	

Exposure

Exposure Factors	Mean	SD
Sea Surface Temperature	2.97	1.11
Sea Surface Salinity	3.01	0.92
Surface pH	3.87	0.32
Mixed layer depth (delta rho = 0.03)	1.12	0.43
Diazotroph integrated prim. prod.	1.3	0.40
Small phyto. integrated prim. prod.	1.22	0.38
Medium phyto. integrated prim. prod.	1.29	0.41
Large phyto. integrated prim. prod.	1.23	0.34
Total Exposure	**High (3.02)**	**0.59**

Vulnerability

Vulnerability	Mean	SD
Total Vulnerability	**Low (3.02)**	**0.59**

Low Moderate High Very High

Figure 1: Summary Sensitivity, Exposure, and Vulnerability Scores and Score Distributions for Butterfish. Exposure and Vulnerability scores are averaged spatially across the species distribution and within stocks if applicable (see Figure 5 and Figure 4). Life History Data Quality (LH DQ) and Standard Deviations (SD) are reported for senstivity and exposure/vulnerability scores, respectively.

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Low	Low	None
Exposure	High	High	None
Vulnerability	Low	Low	None
Directional Effect	Positive	Neutral	Decrease

Stocks

Butterfish is managed as a single stock or stock information was not considered for this iteration of the CVA.

Score Descriptions:

Sensitivity:

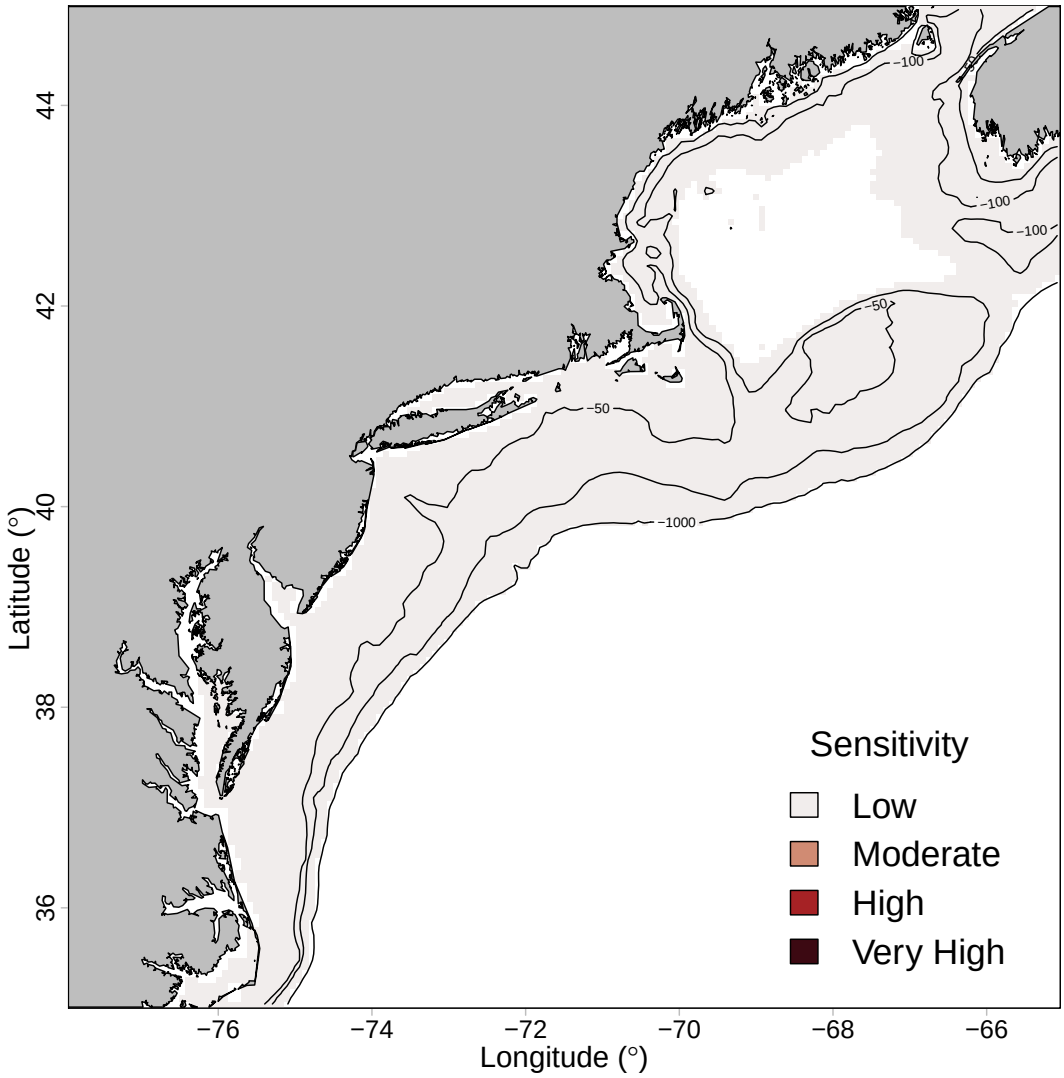


Figure 2: Butterfish Sensitivity on the Northeast Shelf

Butterfish received a **Low** sensitivity score. Certainty was 100% (Figure 1). No attributes had scores greater than 2.5 (Figure 1). The only score greater than 2 was Other Stressors (2.1). Butterfish spawn in estuarine habitats like Long Island Sound and Chesapeake Bay that are impacted by urbanization impacts such as hypoxia, loss of seagrass beds, and changes to freshwater inputs (Guyant et al. 2022).

These results are similar to the 2016 Assessment where all attribute scores were less than 2.5 and the overall sensitivity score was Low (Hare et al. 2016).

Exposure:

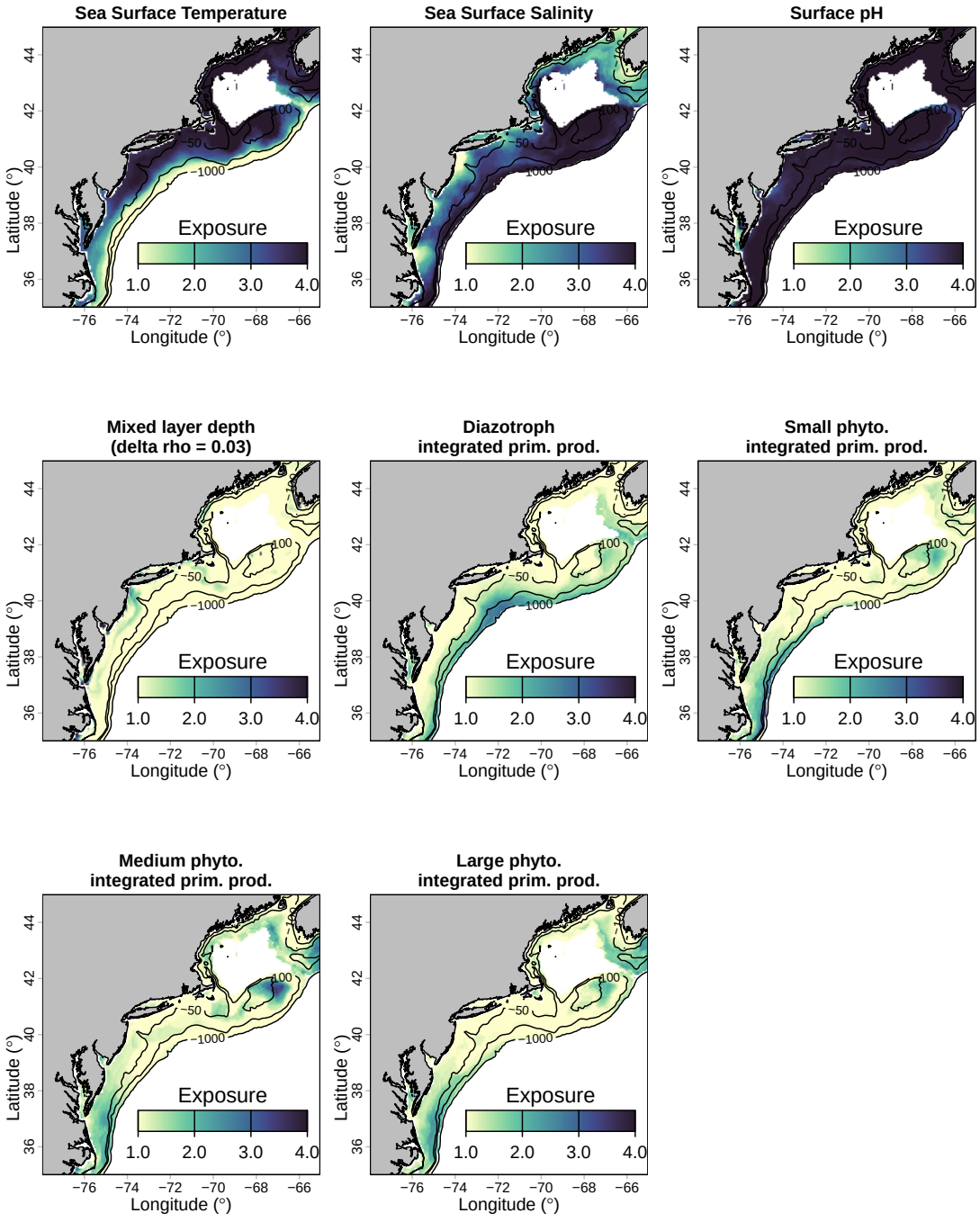


Figure 3: Maps of each Exposure Factor for Butterfish. Exposure was calculated between 2014-2023 and 2025-2035

Mean Exposure for Butterfish was **High** (3.02 +/- 0.59) (Figure 1). Exposure was High throughout most of the species range, with Very High exposure along Georges Bank and Low exposure off the coast

of Chesapeake Bay (Figure 4). These scores were driven by high surface temperature, salinity, and pH exposures (Figure 3).

In the previous Assessment, Butterfish also received a High Exposure score, driven by surface temperature and ocean acidification (Hare et al. 2016).

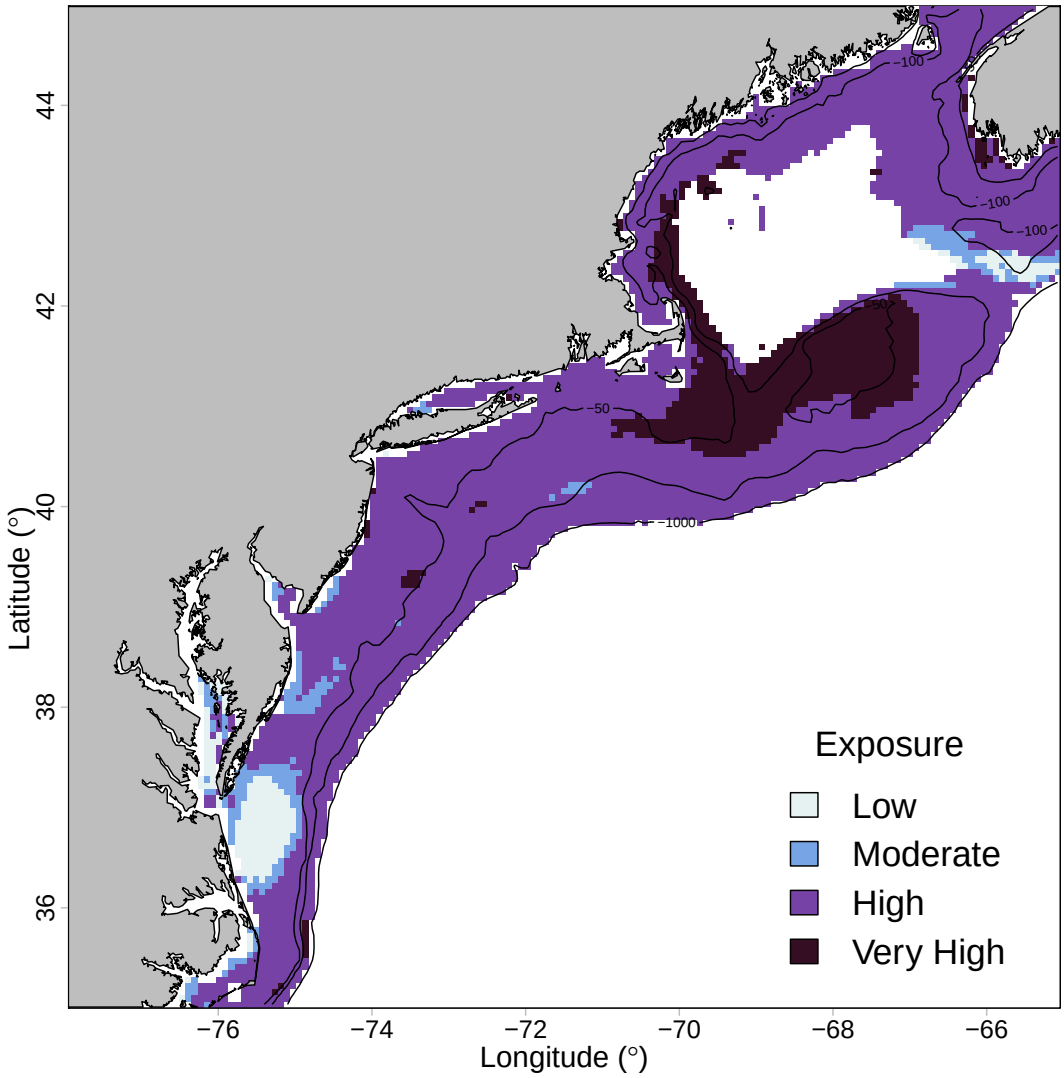


Figure 4: Total Exposure on the Northeast Shelf for Butterfish. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

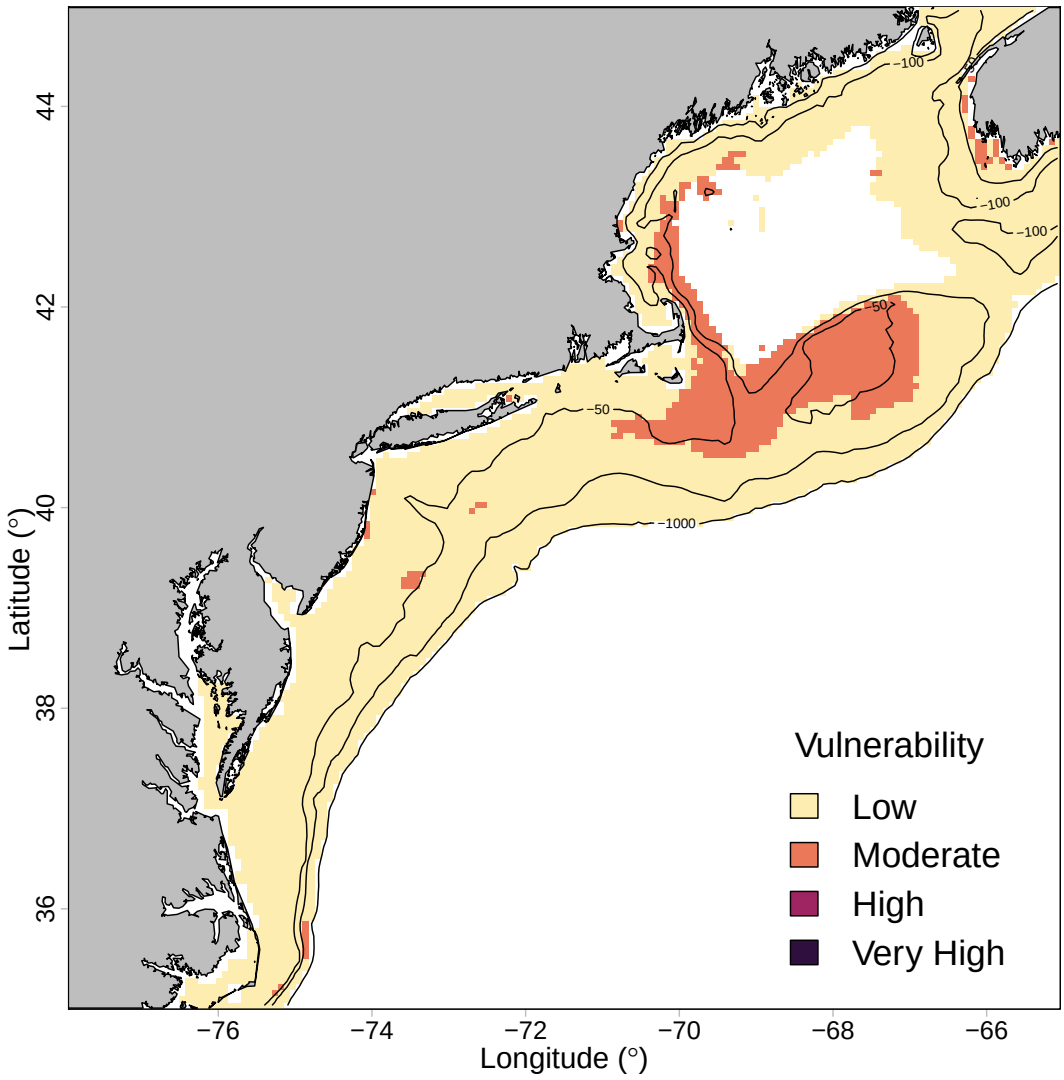


Figure 5: Vulnerability on the Northeast Shelf for Butterfish. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean overall Vulnerability for Butterfish was **Low** (3.02 +/- 0.59), driven by Low Sensitivity (Figure 1). Vulnerability was Low throughout most of the Butterfish's range, but was Moderate along Georges Bank and off the coast of Massachusetts/Cape Cod (Figure 5).

In the previous Assessment, Butterfish also received a Low Vulnerability score (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

95% of data quality scores were greater than or equal to 2, indicating excellent data quality. No data gaps (mean attribute data quality score < 2) were identified (Figure 1).

Model Confidence Score:

The butterfish ensemble model received a mean model confidence score of 1.9 (+/- 0.2), indicating low confidence in the model. Experts stated that the model generally met expectations but tended to over-predict presence. The ensemble model for butterfish had an AUC score of 0.86 indicating good predictive ability.

Additional Metrics

Distribution Shifts:

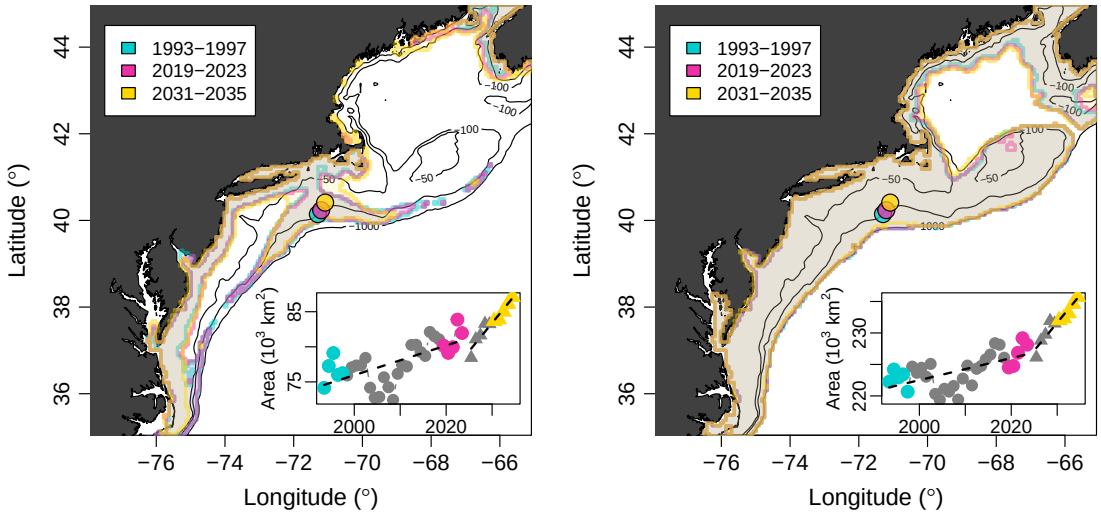


Figure 6: Distribution shifts on the Northeast Shelf for Butterfish. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

Butterfish habitat increased significantly between 1993 and 2023. While butterfish tended to occupy less area on the shelf break, they occupied more area on the shelf itself (Figure 6). The weighted centroid shifted 13 km northeast between 1993 and 2023.

Under near future climate conditions, butterfish habitat is predicted to continue expanding. Offshore habitat use on the shelf break is predicted to decrease, but butterfish are expected to continue to use more continental shelf habitat (Figure 6). The weighted centroid is predicted to shift a further 22 km to the north-northeast.

In the previous assessment, butterfish received a Very High Distributional Vulnerability Rank due to their high mobility and large seasonal migrations (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km^2)	Habitat Total Area (change) (km^2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	77,124	222,896	8.29	9.46
2019-2023	13.0 km NE	81,307 (+4,183)	226,905 (+4,009)	8.29 (0)	9.46 (0)
2031-2035	21.8 km NNE	85,249 (+3,942)	233,659 (+6,754)	8.10 (-0.19)	9.54 (+0.08)

Directional Effect:

The effect of climate change on butterfish was predicted to be **neutral**, with 98% certainty in expert scores. Butterfish have expanded their range northward and into deeper waters (Suca et al. 2021) (Adams 2022). However, reliance on zooplankton and copepod prey in the spring may lead to mismatches that impact recruitment (Suca et al. 2018). In the previous Assessment, experts predicted that butterfish would be positively impacted by climate change (Hare et al. 2016).

Climate Effects on Abundance and Distribution

Butterfish distributions and abundance in the Mid-Atlantic Bight are strongly related to bottom temperature (Kohut et al. 2013). Butterfish have shifted their distribution northward in recent decades. Collie, Wood, and Jeffries (2008) documented an increase in butterfish abundance in Narragansett Bay in the latter half of the 20th century, and they recently have expanded their range into the Gulf of Maine (Suca et al. 2021) (Adams 2022). Their depth distribution has also shifted to inhabiting deeper waters (Suca et al. 2021). An increase in butterfish prevalence in the Gulf of Maine has altered seabird diets, and is replacing preferred prey in common terns (Smith and Craig 2023).

Life History Synopsis

Butterfish distribution ranges from Newfoundland in Canada to Atlantic Florida, but they are primarily found from the Gulf of Maine to Cape Hatteras. They are a schooling fish and shift distribution in response to changing water temperatures: moving north and inshore over the summer to spawn and south and offshore in the winter to seek warmer water (Cross et al. 1999). Butterfish are broadcast spawners and spawn primarily in the evening or at night, however no direct observations have been made (Horn 1970). In general, butterfish spawn starting in May through early fall, peaking in June-July, although timing is dependent on location (Cross et al. 1999). Egg duration and larval growth is temperature dependent (Colton and Honey 1963), and most eggs and larvae prefer temperatures 9-19°C (Cross et al. 1999). Butterfish eggs and larvae are widely distributed throughout their range from spring to fall as spawning occurs over their entire range. Larvae are associated with the Gulf Stream, although it is not clear how far they travel (Morse, Fahay, and Smith 1987). Later life stages (juveniles and adults) can inhabit surface waters to depth on the continental shelf and into coastal bays and estuaries (Schreiber 1973). Survival is reduced in juveniles and adults below 10°C (Cross et al. 1999). Juvenile and adult butterfish feed mainly on planktonic prey including thaliaceans (primarily *Larvacea* and *Hemimyaria*), mollusks (primarily squids), crustaceans, coelenterates, polychaetes, small fishes, and ctenophores (Horn 1970).

Recent diet analysis on 2013-2015 trawl specimens illustrates that spring diets are dominated by *Calanus* copepods, amphipods, larvaceans, and fall diets are dominated by amphipods (Suca et al. 2018). There is little information about larval predation; juvenile and adult butterfish are preyed upon by haddock, silver hake, goosefish, weakfish, bluefish, swordfish, sharks (hammerhead), and longfin inshore squid (Bigelow and Schroeder 1953) (Horn 1970). Butterfish age at maturity is 1-2 years; the maximum lifespan is 6 years although most butterfish live roughly 2-3 years (Horn 1970) (Draganik and Zukowski 1966). In the Northwest Atlantic, butterfish are managed by the Mid-Atlantic Fishery Management Council under the Mackerel, Squid, and Butterfish Fishery Management Plan. As of the 2024 stock assessment, the stock is not overfished and overfishing is not occurring (Northeast Fisheries Science Center (NEFSC) 2024).

Supplemental Scores

Exposure Timeseries

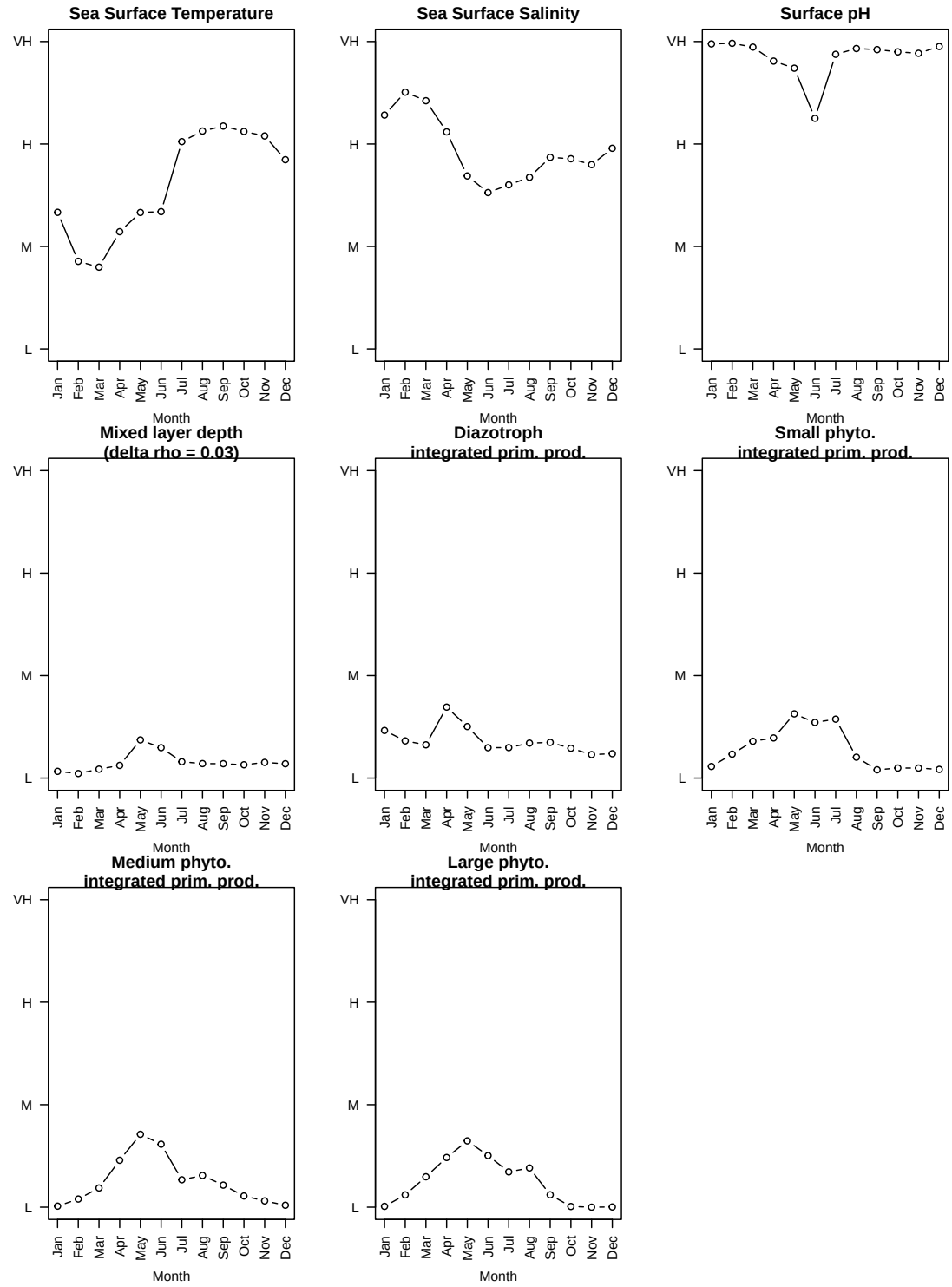


Figure 7: Monthly Timeseries of each Exposure Factor for Butterfish. Exposure was calculated between 2014-2023 and 2025-2035

Exposure was High for Butterfish throughout most of the year. Exposure declined to Moderate in May through June and December (Figure 8). These declines were due to declines in salinity and pH, and sea surface temperatures, respectively (Figure 7).

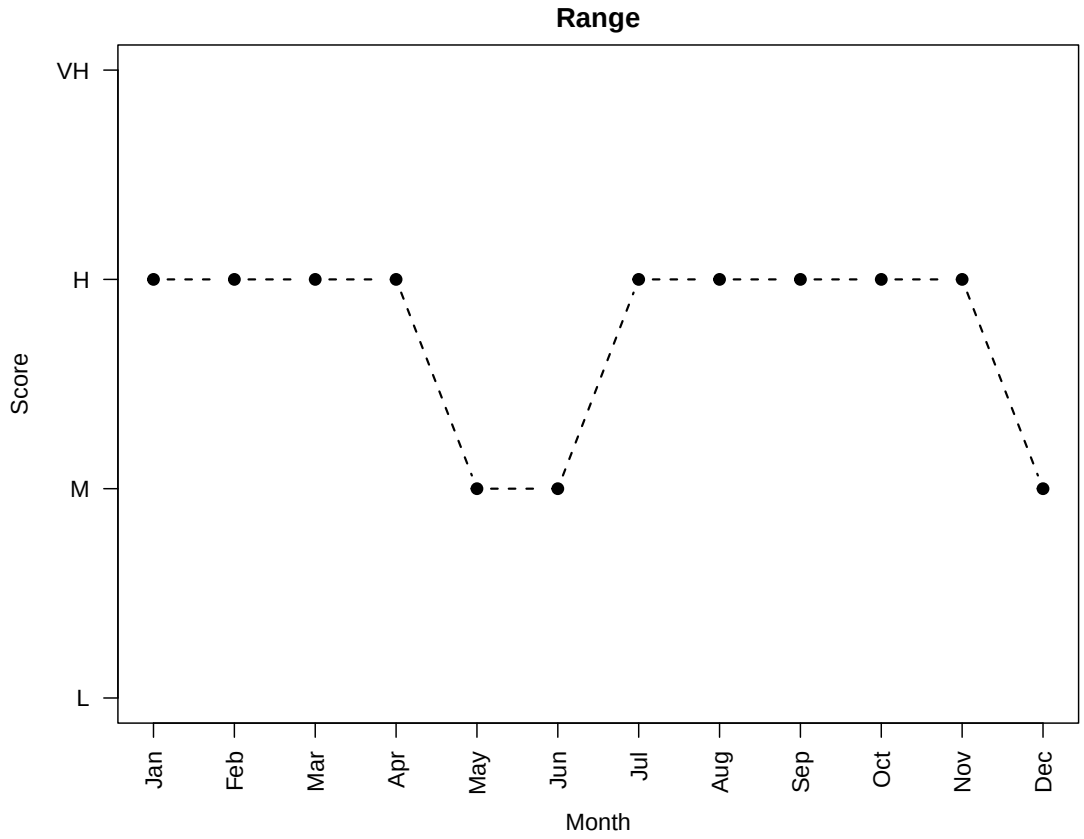


Figure 8: Timeseries of Total Exposure for Butterfish

Exposure due to Significant Factors

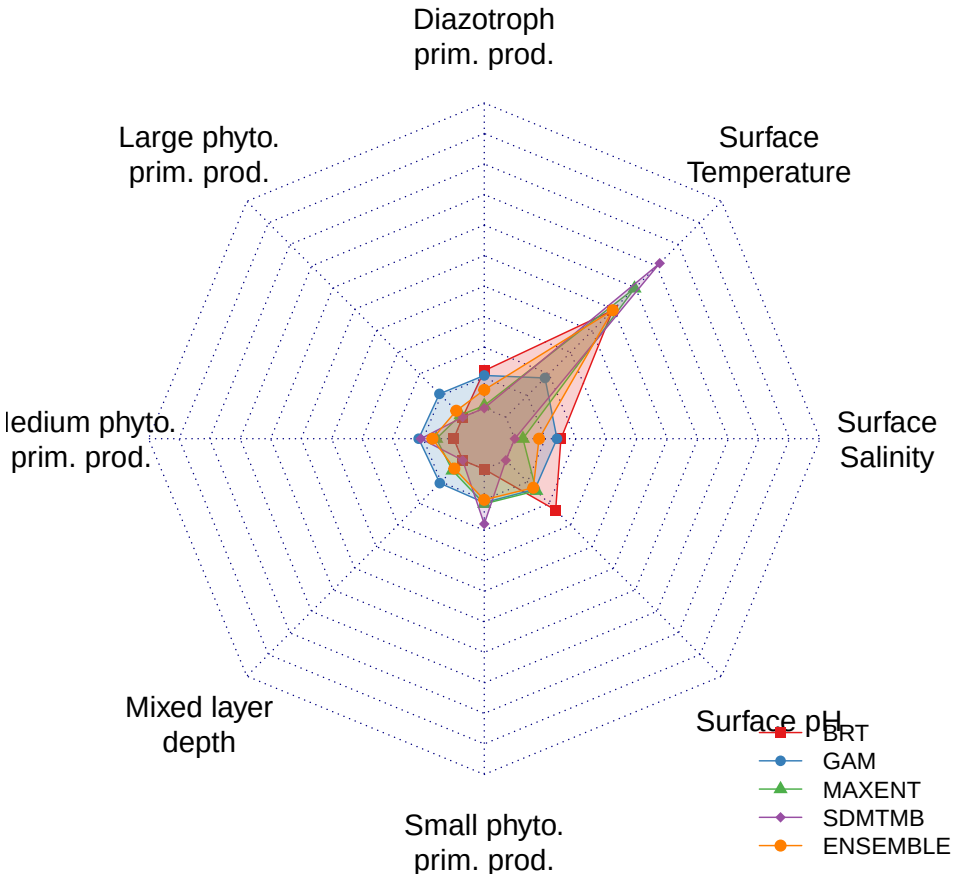


Figure 9: Dynamic Factors driving Butterfish distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Surface temperature, pH, and small phytoplankton productivity were important to species distributions (Figure 9). Temperature is an important driver of Butterfish migrations (Cross et al. 1999). Recent diet analysis shows that *Calanus* copepods, amphipods, and larvaceans are primary prey sources (Suca et al. 2018). When only these important factors were used to calculate exposure, mean exposure decreased to Moderate (2.29 +/- 0.9) (Figure 10;, Figure 11).

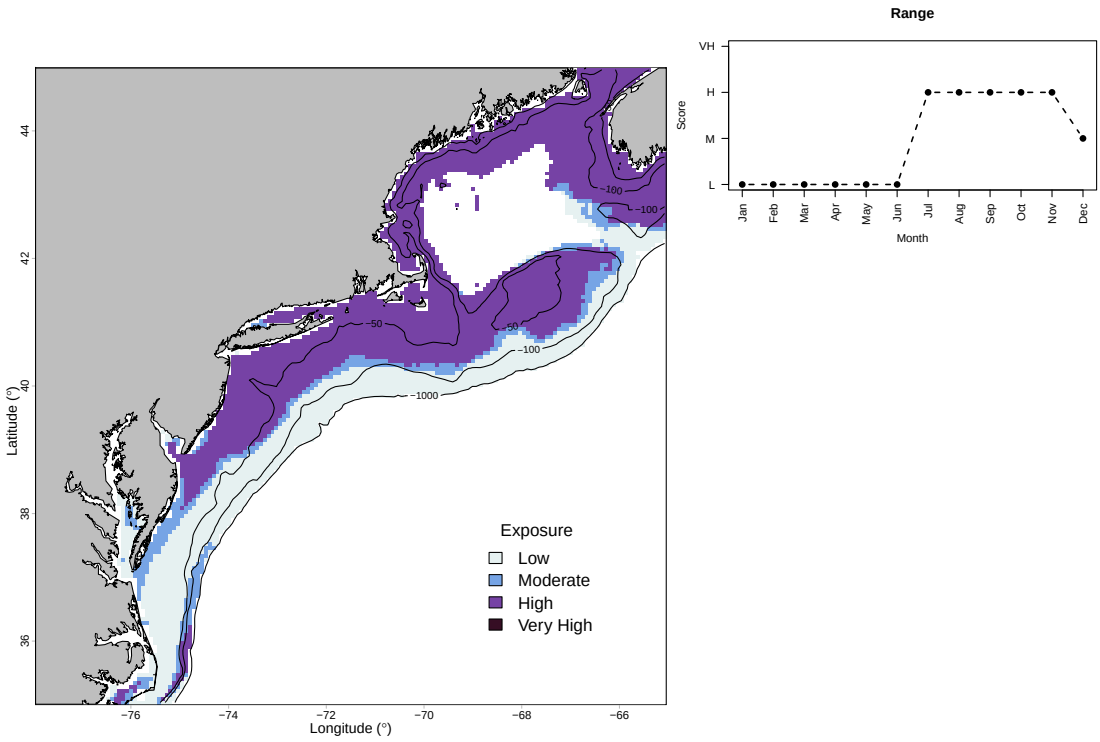


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Butterfish distributions

Sensitivity			
Sensitivity Attributes	**Score**	**LH DQ**	**Tally**
Adult Mobility	1.2	3	
Complexity in Reproductive Strategy	1.76	3	
Dispersal of Early Life Stages	1.28	2.7	
Egg and Larval Survival and Settlement Requirements	1.72	3	
Historical Variability in Temperature	1.2	3	
Juvenile Habitat Requirements	1.32	3	
Other Stressors	2.12	2.8	
Population Growth Rate	1.32	2.85	
Prey Specificity	1.2	3	
Relationship to Calcified Organisms (due to impacts from Ocean Acidification)	1.12	2.4	
Spawning Cycle	1.48	3	
Stock Size/Status	1.04	3	
Total Sensitivity | Certainty	**Low (1) | 1**		

Exposure			
Exposure Factors	**Mean**	**SD**	**Tally**
Sea Surface Temperature	2.97	1.11	
Surface pH	3.87	0.32	
Small phyto. integrated prim. prod.	1.22	0.38	
Total Exposure	**Moderate (2.29)**	**0.90**	

Vulnerability			
Vulnerability	**Mean**	**SD**	**Tally**
Total Vulnerability	**Low (2.29)**	**0.9**	

Low Moderate High Very High

Figure 11: As in Figure 1 for Butterfish, but with only significant Exposure factors and the resulting Vulnerability Scores.

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